

Yujia (Helen) Zheng

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Master of Science in Software Engineering - Scalable Systems (Accelerated Program)

May 2026

Bachelor of Science in Information Systems, Additional major in Computer Science

May 2025

Coursework: Distributed Systems | Web Application | Computer Systems | API Design | Agile Methods | Quality Assurance | Database Development | AI Theory & Practice | Foundations of Software Engineering | Parallel and Sequential Algorithms | Object-oriented Programming | Imperative Programming

SKILLS

Backend / Systems: Java, Python, C, Spring Boot, Node.js, RESTful APIs, RPC, MySQL, PostgreSQL, MongoDB, Redis, Kafka

Web / Frontend: JavaScript, TypeScript, React, HTML, CSS, Client-Server Communication, HTTP, TCP/IP

Tools / Platforms: Git, GitHub Actions, JUnit, PyTest, Postman, CI/CD, Docker, JWT Authentication, AWS, CloudWatch

EXPERIENCES

Surefront

Remote

Software Engineering Contractor

Aug – Dec 2025

- Collaboratively designed and developed an **AI-powered MCP chatbot**, integrating **Django REST APIs** with an **AWS-based infrastructure (EC2, Lambda, S3, DynamoDB, PostgreSQL, Redis)**
- Built and optimized backend modules for **session management**, **message orchestration**, and **data persistence**, enhancing performance and scalability through **Redis caching** and efficient **API design**
- Partnered with frontend and platform engineers to conduct **unit and integration testing**, ensuring **API reliability** and **multi-session consistency**, and deployed via **AWS CloudFront** and **Elastic Load Balancing** for scalability

Shanghai Magus Technology Co., Ltd.

Shanghai, China

Software Engineering Intern

May – Aug 2025

- Implemented a secure job management system with **Java** and **Spring Boot RESTful APIs**, integrating **JWT authentication** for role-based access control and data security
- Designed and deployed backend services with **Spring Boot**, **JPA** and **MySQL**, streamlining job creation, retrieval, and deletion workflows, reducing manual overhead by **50%**, and improving task ownership accuracy
- Developed a **React-based dashboard** supporting real-time job creation and configuration with customizable inputs and live status tracking, improving workflow visibility and decision-making efficiency

RESEARCH & PROJECTS

Santorini Full-Stack Application | Undergraduate Project

Jan – May 2025

- Designed and built a full-stack Santorini board game with **Java RESTful APIs** and a **React frontend**, delivering interactive multiplayer gameplay with real-time state synchronization via a custom **JSON** client-server protocol
- Implemented **JUnit** test suites (90% coverage) and deployed **GitHub Actions CI/CD** pipeline to automate testing

Server and Client of an RPC System | Undergraduate Project

Jan – Feb 2024

- Built a file access **RPC system** in **C**, including a server for file services and a client stub library, enabling distributed operations such as open, read, and write
- Designed a custom **serialization protocol** to ensure efficient, reliable communication between client and server

The Technology for Effective and Efficient Learning Lab | Undergraduate Research

May – Aug 2022

- Collaborated in a research team to develop an automated multiple-choice question generation system using **Python** and **GPT-4**, enabling large-scale creation of programming education questions
- Co-authored and published the peer-reviewed paper [A Comparative Study of AI-Generated \(GPT-4\) and Human-crafted MCQs in Programming Education](#), which contributed novel insights into AI-assisted assessment design

OpenStack Manila Project | Undergraduate Project with Red Hat

May – Aug 2022

- Added **Network Subnet Metadata**, **Soft-delete**, and **Restore APIs** in **Python** to enhance OpenStack Manila's cloud-based file storage features
- Followed open-source workflows using **Gerrit** and collaborated with **Red Hat engineers** on code reviews, also writing a **user guide** to boost adoption